Lexical Task Heads on Simple Tasks

Schematic project summary

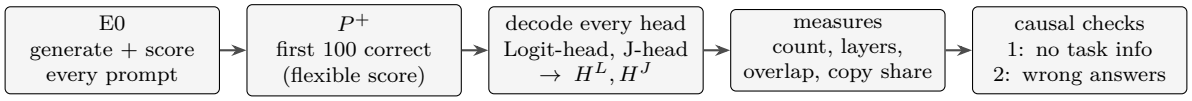
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Marks: ✓ works / supports the hypothesis, ✗ does not, ∼ mixed or task-dependent. Full methods and tables: `proposal-simple.tex`. The complex tasks are summarised in `summary-complex.tex`.

1 Setup in one page

Question	Does the lexical-task-head method of Yang et al. (2026) reproduce on our setup, does Head-JLens (J-head) find heads the per-head logit lens (Logit-head) misses, and which heads should be kept when a task is asked for in several wordings?
Model	Llama-3.1-8B-Instruct (bf16; only model with released Head-JLens matrices).
Readouts	Logit-head: head write $c_{l,h}$ decoded by the unembedding. J-head: head output $z_{l,h}$ mapped through the fitted Jacobian $J_{l,h}$, then decoded.
Simple tasks	14 Todd et al. tasks (antonym, capitals, translation, ...). Rule style: the first 5 of Yang's 6 instruction templates (A–E, incl. the cloze form); example style: 1/5/10/20/30 shots; zero-shot prompt. Raw text, no chat template.
Head criterion	Head is a lexical task head on a prompt if ≥ 1 of its top-10 tokens is in the task's term list; for the style if this holds on $\geq 10\%$ of P^+ (first 100 correct prompts of the most accurate wording).
Term lists	Yang et al.'s (2–22 token ids). Since 11-05 also lists built from the words all heads decode (J-head; E3 below).
Causal check 1	Yang App. G.1: patch the mean head output ($\times\alpha$) at the last token of prompts *without* task info; control = mean of another task at the same heads.
Causal check 2	Yang Sec. 2.4.1/App. H.1.3 in their released script's setting (1-shot): patch prompts that *give* the task but were answered wrongly (baseline 0).
Head sets patched	H^L, H^J, and J-only $H^J \setminus H^L$.



2 Experiments and outcomes

Experiment	Design	Outcome (short)	
E0 — behavioural baseline			
Simple tasks	14 tasks × 11 prompts, every item.	All tasks well above zero-shot; accuracy rises 1→5 shots then flat. Template spread largest on translation, next/prev-item, park–country, synonym.	✓
Scoring audit	Every reply saved; strict (first token) vs flexible.	Translation spread mostly quotes/capitals; first-token score marks any number correct.	✓

E1 — simple tasks (14 tasks × rule/example × 2 readouts)

Experiment	Design	Outcome (short)	
Head sets	100 prompts per style; most accurate template / 5-shot.	Logit-head finds heads on every task. J-head finds more on 23/28 pairs; its extra heads sit in layers 10–19 (Logit-head mostly ≥20). J-head contains all of Logit-head only on 4/28. Example heads mostly inside rule heads (70% shared by Yang's ratio; they report 73%). Counts within 1 head of Yang's Figs. 79/88 in their code's setting; below their Figs. 28/20.	✓
Check 1 (zero-shot)	Means per wording, 15 α values, control mean.	Logit-head: gain ≥10 items on capitals, landmark, park, product (+ currency, next-item, plural with examples). Nothing on antonym, synonym, present–past, prev-item, translations. Best around $\alpha = 4$. Control mostly low (not on park–country, next-item).	~
Check 1, J-head	same	J-head ≥ Logit-head on most tasks; J-only heads have effects of their own.	✓
Cross-style means	Rule heads × example means and vice versa.	Same tasks gain, often as much or more.	✓
Check 2 (1-shot wrong)	Yang's released script, no control.	Gains on all 14 tasks; reproduces Yang's Figs. 57/62 task order. J-head far stronger on next/prev-item with examples.	✓
E1b — heads from several templates (rule style, simple tasks)			
$k = 5$ (all templates)	Heads passing under all 5 templates; both checks.	Better on fact tasks (capitals, park, product) only; worse elsewhere. Beats E1 on ≤5/14 tasks (≤3 by >3 items).	✗
$k = 1$ (union)	Exploratory.	Helps J-head on present–past (0→16 items), synonym, prev-item, control flat. Hurts Logit-head on fact tasks.	~
$k = 3$	Exploratory.	Mostly the E1 heads; behaves like E1.	–
$k = 1$, shared	J-head union, kept only heads in the union sets of ≥ 2 tasks (keeps 15–62 of 19–74 heads per task); both checks.	**Not successful.** Removes most of the union's gains: present–past 16→1 items (check 1), 0.64→0.45 (check 2); synonym 0.21→0.13. Only country–capital rises (0.60→0.69). Shared heads do not carry the task better.	✗
$k = 1$, 4 present–past heads	The 4 heads only present–past's union set contains (L19H10, L19H24, L20H28, L21H9; none in E1); both checks, present–past only.	Check 1: no effect alone (0.01); needs the other 15 heads too. Check 2: 0.63, as the whole 19-head union (0.64); rises with α up to 10, real past forms (60/61 distinct replies); $\alpha = 0$ gives 0.02.	~
Replies, present–past	Each patched reply sorted by kind, per head set and α (J-head, rule means; notebook below).	Shared heads make the model *name* the task ("past tense of assess": 65% of replies in check 1, 73% in check 2 at $\alpha = 10$) without doing it. The 4 unshared heads make it give *a form of the input*: from a bare prompt it repeats the verb (44%), when the prompt shows the task they fix the inflection. Together (union) they give the answer. One task; heuristic reply kinds.	–
Replies, all tasks	Same, every task.	Many patched replies contain the expected answer after a prefix the flexible score rejects ("Capital – Tbilisi", "plural needles", "Country – Russia, . . . "): 19–33% of E1 J-head replies at $\alpha = 4$ on capitals, landmark, antonym, synonym, singular–plural. Not all are answers ("opposite of solid" for *hollow*).	–
Retrieval heads	Yang et al.'s retrieval heads (their code): Definition 1 with the prompt's own answer token as target, on E1's identification prompts; overlap with every lexical set, against layer-matched chance.	Mostly distinct from lexical task heads: 2–9% of each set's heads are retrieval heads, at chance under Logit-head (0.6–1.2×), slightly above under J-head (1.1–1.7×, highest for $k = 5$). None on the translation tasks. Of the 4 present–past unshared heads only L21H9 is one (15% of prompts).	–
4 heads split	Present–past, both checks: L21H9 (the one retrieval head of the 4) alone, and L19H10, L19H24, L20H28 alone.	L21H9 carries the effect: check 2 0.60 alone (4 heads 0.63, 3 heads 0.24); in check 1 it alone makes the model repeat the verb (44% at $\alpha = 10$), as the 4 heads do, and the 3 heads do nothing. A retrieval head that brings the input word forward; the inflection comes from the prompt's example (check 2) or the shared heads (check 1).	~

Experiment	Design	Outcome (short)	
$k = 1$ without retrieval heads	$k = 1$ sets (L and J) minus every retrieval head of the task (any template, either readout): 51/389 L and 84/672 J heads removed, none on translations; both checks, all tasks.	Mostly small changes, but task-dependent. Landmark–country collapses (check 1: L 0.72→0.02, J 0.75→0.13; J replies still name the country, not first). Present–past J loses its gains (L21H9 removed). Gains on synonym J (0.21→0.32), product–company J (0.35→0.48, control 0.10) and prev-item J check 2 (0.12→0.45, at α 9–10, no constant answer). Park–country L: control now above task. Unchanged sets reproduce $k = 1$ exactly. Broad removal: retrieval heads of E1's template and the same readout only would remove 12 L and 51 J heads.	∼
Equal-size sets	Top $N \in \{5, 10, 20\}$ heads by match rate (ties: term count) under each readout and rule (E1, $k = 1/3/5$), only heads matching ≥ 1 prompt (37/420 sets not formed); both checks, own-style means, 9 α ($-10, -4, 2$–$6, 8, 10$); threshold sets compared on the same 9.	**Readouts: J-head's lead is not set size.** Of 25 threshold-set J leads ≥ 0.05, 16 hold at every N, none reverse; none of Logit-head's 4 leads holds (country–capital, present–past reverse). **Rules: mostly size.** Most E1-vs-k differences flip with N; consensus ($k = 5$) below E1 in check 2 (J: antonym, singular–plural, next-item, German). No steady gain with N. Present–past J example: threshold (4 heads) 0.00, top 5 0.87 by adding L15H1 (rate 0.07), control 0; only sets with L15H1 move check 1. L15H1 alone (example means): 0.80 / 0.76 in checks 1 / 2 (top 5: 0.87 / 0.80; the 4 without it 0.00 / 0.57), control 0, none with rule means; J-head reads past participles from it. Control high on park–country (title starting with the country), singular–plural J ex $N = 5$, next-item J ex $N = 20$.	∼
Davidson FV	Davidson et al.'s (2025) instruction function vector, 4 tasks (landmark, product, English–French, present–past): heads ranked by the gain in answer probability when set to their mean on prompts whose instruction is replaced (equiprobable tokens, WikiText, other tasks' instructions), top 20 over the 4 tasks; FV = sum of their mean outputs (inst1–4, 100 prompts), added after layer 11 at the last prompt token; our check 1, 15 α ($\alpha = 1$ is theirs), control.	Peak at $\alpha = 1$ everywhere, nothing at $\lvert\alpha\rvert \geq 3$. Landmark 0.85, product 0.64 (controls 0.06 / 0.12; our best sets 0.79 / 0.46 at $\alpha \approx 3$–4); English–French 0.09 (control 0.03; ours $\leq$0.02); present–past 0.02 (repeats the verb, as our sets). FV heads in layers 13–17 mostly, 0–5 shared with each E1 set. English–French: 8 of the 9 hits are the English input copied where the French differs only in accents. **80 heads** (next 60 by the same score; FV 25–30% longer, cosine 0.88–0.91 with the 20-head FV): still best at $\alpha = 1$; landmark 0.96, product 0.61, English–French 0.19 (11 real translations), present–past 0.07 (real past forms); controls $\leq$0.10. **Set size** (5–1,024 heads, same ranking): no single best. Facts: 5–10 heads at α 3–4 best (landmark 0.95, product 0.87; product control 0.22), at $\alpha = 1$ 40–80 heads (0.95–0.96 / 0.61). English–French: 40 heads, $\alpha = 1$, 0.28 (0.20 real translations). Present–past: 320 heads, $\alpha = 1$, 0.18 (real past forms). All 1,024 heads: facts break (0.01). Size chosen on the test items.	∼
Davidson FV, 14 tasks	As above, heads chosen over all 14 tasks (30 heads; next_item: 12 and 28 correct prompts on two templates, so fewer for its mean); our check 1, all 14 tasks.	Best at $\alpha = 1$ nearly everywhere; FV $\geq$ our best E1 set on 11/14 tasks, controls low ($\leq$0.12). Large gains where our sets do little: singular–plural 0.60 (ours $\leq$0.03), antonym 0.26 ($\leq$0.04), park–country 0.89 (0.39–0.48, our controls 0.29–0.32), English–Spanish 0.10 (real translations). Also above: capitals 0.83 (0.70), landmark 0.89, product 0.64, currency 0.13. Nothing on present–past, prev-item, synonym, English–German (0.01–0.04); English–French 0.09 = copied inputs (accents ignored). Next-item 0.10 vs J 0.17 (J control 0.35).	∼

Experiment	Design	Outcome (short)	
FV heads read	The 30 FV heads (14 tasks), top 10 at the last token: their outputs on correct inst1–4 prompts; overlap with CC-pooled; and the difference mean output (correct prompts) − output on each of the 60 corrupted prompts of the FV score.	Logit-head: noise but 4 heads. J-head, ∼20 readable: answer category (L16H25, L17H4: *cities, currencies, countries, companies*), task domain (L14H13, L15H27, L15H21), relation (L13H22: *opposite, replacement, twice, earlier, next, preceding*), example answers (L17H5, L21H2); ∼1/3 punctuation or noise; answer share ≤0.10. CC-pooled holds 3–11 of the 30 per task (layer-matched chance 4–6); 10 in no CC set (L17H5, format, noise); 34/130 memberships by common words only. Difference: same words for category, domain, relation heads; content under constant punctuation (L14H29 *counterpart, equivalent, translated*; L15H22 *individual, today, previous*); late heads show the item's input. Answer share of the difference finds 5/30 FV heads.	–
Lens FV score	Davidson's head score estimated without patching: per head, the change (task mean − output on each of the 60 corrupted prompts) through J-head (or Logit-head) and the final norm linearised at the prompt, projected on the prompt's own answer token (logit, log-prob, prob); the measured score recomputed on the same prompts.	Checks: means = saved FV, measured = saved score (≤0.001), norm step = finite difference. Over 14 tasks, J-head: Spearman 0.31–0.35 with the measured score (Logit-head 0.14–0.24); 7–8 of the 30 FV heads in its top 30 (Logit-head 4–5, chance 0.9). Finds the late FV heads (L31H2, L31H14, L31H22, L18H29, L17H5, L16H25, L21H2), misses the mid-layer word heads (L15H14 rank ∼950, L13H22 ∼200); its own top 30 mostly layers 24–31.	∼
FV generic heads	FV heads whose J-head tokens are alike across the 14 tasks (mean cosine of token counts), against the 231 equally generic heads of layers 13–31: attention at the last token (split by prompt part), mean output per task (J-head-transported), change by the instruction, per-task score.	6–7 generic: L14H29, L15H5, L14H11 (*...\n, :\n, ;\n*), L13H31, L18H22, L31H14 (Logit-head: *, (a in*), L15H7 (*instead, named, Wikipedia*). Near-constant output (cosine 0.80) and small instruction change (0.17) are shared by the equally generic heads (0.78, 0.18); what differs is attention off BOS (0.25 vs 0.88). L14H11, L15H5, L13H31 put 40–62% on the answer slot (newline, *A*, final colon), L31H14 99% on the last token; the task FV heads read task words (0.21–0.32) and the input (0.14–0.29). Generic decoding with attention off BOS: 9/608 heads, 5 FV (score ranks 9–27).	–
FV split	Davidson's 30-head FV in our check 1, split: the 6 generic heads (generic6), the other 24 (task24), and the 30 minus 6 random non-generic heads (3 draws); $\alpha \in \{0, 1, 2, 3, 5, 8\}$; reply length and line-break stops; all30 = saved run (7/168 rows differ by ≤0.03, batching).	generic6 alone: nothing (0.03). **task24 beats the full FV**: 0.58 vs 0.33 (best α per task), ≥ all30 on all 14, best at α 2–3 (0.54 at fixed $\alpha = 3$); translations 0.39–0.70 (FV ≤0.10), antonym 0.61, synonym 0.35. Dropping 6 random task heads: 0.25–0.30, still peak at $\alpha = 1$. The generic part is most of the FV's norm (9–15 of 11–16), the same direction on every task (cosine 0.96), orthogonal to the task part; scaled up it turns replies into *://*, *.swing*, *1. 2. 3.* on every task.	✓

E3 — head sets from decoded words (J-head only, rule style; check 1 only, rule means averaged over the 5 templates, control mean, $\alpha \in \{1, 2, 4, 8\}$, best α reported. Numbers are means over the 14 tasks; E1 J-head on the same α: 0.19.) Words: a decoded top-10 token, stripped and lowercased, kept if alphabetic, ≥ 3 letters, not a stop word, common English (`wordfreq` Zipf ≥ 3); *share* = fraction of (prompt, head) pairs over all 1,024 heads whose top-10 holds it; *task words* = share in the task − mean share in the other 13; *common words* = highest minimum share over the 14 tasks. Heads: Definition 1 with the new list ($k = 10$, $\rho = 0.10$) unless stated. Only P^+ (correct) prompts.

Top decoded words	Top 40 words per task over all heads, no baseline.	Task names and answer types lead (*opposite*, *city*, *currency*, *French*); generic words on every task (*what*, *which*, *maybe*, *https*). Yang's list words only partly among them.	–
Common + task, 4 tasks	20 common and 20 task words from the best template; heads on the best template; both readouts; antonym, present–past, capitals, product.	Task list J best (capitals 0.59, antonym 0.22); common lists ≈ 0 except product under Logit-head (0.32).	∼
All 14 tasks	Same, words counted on all 5 templates.	Task J 0.10, common J 0.07: below E1.	✗
List length	5, 10, 15, 20 words.	Shorter is better: task J 0.22 (5 words) → 0.10 (20). Common ≤ 0.07.	∼

Experiment	Design	Outcome (short)	
Heads on pooled prompts	Definition 1 on the ≈500 prompts of the 5 templates pooled instead of the best template; also Yang's list.	Helps every list: task J 5 words 0.25; common J 0.11–0.17 (synonym 0.58 with 10 common words, control high); Yang's list pooled 0.21.	∼
Combined 5+5 (2×2)	5 task + 5 common words; words and heads each from the best template or pooled.	**Both pooled = CC-pooled**: 0.45, 95–137 heads, above E1 on all 14 tasks; carries tasks E1 does not (antonym 0.54, singular–plural 0.64, next-item 0.43, translations 0.25–0.32). Present–past 0.03. Controls: synonym 0.26, park 0.17, singular–plural 0.12, next-item 0.12. Words pooled, heads best 0.30; words best, heads pooled 0.31; both best 0.18.	✓
Random sets	Best set per task so far vs 5 layer-matched random sets of its size.	Best 0.49, random 0.03–0.07. Single draws high on antonym (0.45) and landmark (0.42); product 0.16–0.39 on all draws.	✓
Strict consensus	5+5 and 10+10 lists, head must pass on all 5 templates, top-5 tokens, $\rho = 0.15$ (run at $\rho = 0.05$ stopped); also its top 20 by rate.	0.15–0.20. Best only on country–capital (0.78).	✗
Neutral cut (CC-pooled-clean)	J-head readings on 1,000 WikiText-103 passages (lengths matched); the 50 words heads read most there removed before choosing task and common words; 5+5 and 10+10, pooled.	5+5: 0.38, 10+10: 0.33. Above CC-pooled on 8/14 (facts, translations, morphology: landmark 0.97, French 0.40, German 0.49, singular–plural 0.72, present–past 0.14). Loses antonym, synonym, next/prev-item: they need the generic words. New common words: *named, therefore, question, answer, very.*	∼
Task minus WikiText	Top 10 task words after the cut; no cross-task contrast, no common words.	0.21. Landmark 0.00 (patched replies start "Country"), translations name the language.	✗
Group contrast	5 groups (word relations, morphology, facts, sequences, translations); task words = share − max share of the other tasks of its group; + 5 top WikiText words (*which, mph, http, his, what*).	0.32. Best antonym of all (0.70). Landmark and park collapse (0.01 / 0.08): *country* is shared within the group and removed.	∼
Match-rate ranking	CC-pooled heads ranked by pooled match rate; top 1, 2, 5, 10, 20, 40; layer-matched random at 10, 20.	Small sets work only on capitals (L16H25 alone 0.88) and landmark (40 heads 0.93). Means: 5 heads 0.19, 20 heads 0.21, 40 heads 0.25.	∼
Relation heads	Head's top-10 holds both a term-list token and a token of the prompt's own query.	Rare: $\rho \geq 0.10$ for ≤ 7 heads per task, none on 10 tasks. Not patched. L16H25 decodes *city*, not the relation.	–
Attend + decode	L16H25 pattern: at the last token it attends to *capital*, *country* in the instruction and decodes *city*. Score = (share of last-token attention on task words in the instruction, outside the query) × (share of prompts with a task word in the J-head top 10); task words = Yang's list ∪ 10 top task words; eager attention, pooled prompts. Top 1, 2, 3, 5, 10, 20; layer-matched random at 5, 10.	Few heads carry fact tasks and plurals: capitals 0.88 (L16H25 alone), landmark 0.79 (5 heads), product–company 0.72 (20 heads, control 0.09; best of all), singular–plural 0.68 (20). Antonym 0.25 (5). Nothing on translations, next/prev-item, present–past. Park: control nearly as high. L16H25 ranks first on all 5 fact tasks; pattern in layers 13–17. Means: 5 heads 0.21, 20 heads 0.25 (match rate 0.19 / 0.21); random 0.04.	∼

E4 — held-out test of the E3 sets (J-head, rule style, E3 designs frozen. Items of each task split in two halves; words, heads, rankings and means chosen on one, patched on the other, both ways. Both checks, control mean in both; 10 α (the 9 of the equal-size sets + $\alpha = 1$); α cross-fitted: the best on the other fold. Accuracies only, no intervals. Numbers: means over the 14 tasks, check 1 unless stated.)

CC vs E1	CC-pooled (97–154 heads) and the E1 set (7–50), both rechosen on each half.	CC 0.44 vs E1 0.20 (E3: 0.45 vs 0.19). Above E1 on 13/14 (not product–company); still carries antonym, singular–plural, translations; next-item lead small; present–past 0.02.	✓
Random	5 layer-matched draws per size, at the CC set's α.	0.02 at every size (product–company 0.15); check 2: 0.11–0.13.	✓

Experiment	Design	Outcome (short)	
Decoded vs causal	Same number of heads ranked by Davidson's FV score on the choosing half: FV-20, FV-40, FV-N_t.	**FV better at every size:** N_t 0.59 vs CC 0.44; 20: 0.38 vs 0.22; 40: 0.43 vs 0.26. CC ahead only on prev-item and translations at N_t (German 0.39 vs 0.01). FV carries present–past (0.85). FV best at α 1–2, CC 2–3. Davidson's own 30-head FV ($\alpha = 1$): 0.39.	✗
Check 2	1-shot prompts answered wrongly.	Sets close: E1 0.31, CC 0.33, FV-N_t 0.39; controls 0.12–0.13, as large as the task mean on next/prev-item; 11–34 items per fold on 5 tasks.	∼
Secondary	Fold 1 only, best α: CC-clean, attend + decode 20/40, FV-univ 20/40 (FV heads ranked over all 14 tasks, Davidson's choice).	CC-clean 0.39, A+D 0.28 (product–company 0.80), FV-univ-20 0.63, highest of all sets.	–
E4b — more wordings, held-out wordings (design in `proposal-simple`, Sec. 16.3)			
New wordings	30 per task: 3 forms (cloze, Q–A, label) × task named / not named; GPT-5.5, script checks, hand check; then heads from $m \in \{1, 5, 11, 23\}$ wordings at a fixed prompt count, tested on held-out wordings.	84 cells × 5 generated (+ 2–5 spares each). **Pending:** hand check; nothing run yet.	–
E6 — heads from what the instruction adds (J-head, rule style; E4's split, CC rule, patching and cross-fit, frozen)			
Difference readings	Per E4 identification prompt (inst1–4; inst0 has no instruction), clean z minus the mean of two corrupted versions, instruction replaced by Davidson et al.'s length- and log-prob-matched text (equiprobable tokens, WikiText; other-task instructions left out: related tasks share processes), top 10 under J-head. Sets: dcc (CC rule), dccT (5 task words only), top 20/40, answer heads removed (A). E4's CC-20 re-patched = E4 exactly.	The common words do not cancel (*what, https, http, mph*): dcc ≈ CC size (84–128 heads), dccT 21–68. Check 1: dcc 0.18, dccT 0.16, dcc20/dccT20 0.22, dcc40 0.22, dccT40 0.20; CC 0.44, CC-20 0.22, CC-40 0.26, FV-N_t 0.59, random ≤0.04. Ahead only on capitals (0.80–0.84 vs CC-20 0.77) and product (0.60–0.62 vs 0.51); lose antonym, synonym, translations, sequences, singular–plural. dcc keeps ∼half of CC's heads; the dropped ones decode the words with and without the instruction. Check 2: all 0.32–0.34 (CC 0.33). Answer removal: little.	✗
E7 — most active words, then an attention filter (J-head, rule style; E4's split, word rules, CC head rule, patching and cross-fit, frozen. Motivation: word-decoding heads outside the FV set attend to BOS (∼0.80), FV task heads to the prompt (0.25–0.39).)			
Lists and sets	Raw list: the task's 10 most active words (share over the 500 pooled prompts of the choosing half), no cross-task contrast. Clean list: the same after removing the 50 top WikiText words. Heads: CC rule (≥10% of the 500). Filter: drop heads whose mean last-token attention on BOS over the task's prompts is > 0.5.	Raw lists ≈ half task words, half generic (*what, which* on all 14, *http* on 7); clean lists task words only. Folds agree on 9–10/10 words (next-item 4/10). Sets: raw 72–135 heads, clean 58–101; the filter keeps about a third or less (raw 17–43, clean 20–39; median BOS of the unfiltered sets 0.57–0.77).	–
Check 1	Raw, raw + filter, clean, clean + filter; random-drop (as many heads as the filtered set, drawn from the unfiltered set) and layer-matched random, 5 draws each, at the filtered set's α.	Raw 0.37 → **raw + filter 0.40** with 32 heads, between FV-20 (0.38) and FV-40 (0.43) and above them on 8–9/14 (CC 0.44 with ∼120, CC-20 0.22, CC-40 0.26, FV-N_t 0.59); clean 0.18 → 0.20. **Random-drop 0.09–0.10**, layer-matched 0.01–0.02: the filter keeps the heads that carry the task. Filter better on 10/14 for both lists; loses antonym (0.62 → 0.46) and Spanish (0.44 → 0.12). Raw + filter above CC on 9/14 (singular–plural 0.88, capitals 0.87, German 0.54, product 0.46, present–past 0.21 vs 0.02) and above FV-N_t on prev-item and translations; landmark collapses on every E7 set (0.00–0.05 vs CC 0.92; replies *Country: Germany*, *A Journey Through Time*). Controls ≤0.04.	✓

Experiment	Design	Outcome (short)	
Check 2	The four main sets, control mean.	Raw 0.31, raw + filter 0.34, clean 0.29, clean + filter 0.32 (CC 0.33, FV-N_t 0.39). Prev-item raw + filter: control 0.49 above task 0.15.	∼
E8 — every combination of five head filters (J-head, rule style; E4's split and protocol, $\alpha \in \{1, 2\}$ only, cross-fitted; references re-cross-fitted over α 1–2: CC 0.40, FV-20 0.33, E7 raw + filter 0.31, FV-N_t 0.59. Report: `paper/e8-filter-combos`.)			
Filters and sets	D: J-head top 10 holds a word of the task's 30 most active decoded words on ≥10% of prompts (E7 word rules; tokens with leading-space capitalised/upper forms added); B: BOS attention ≤0.5; T: attention on the instruction's task words (Yang terms + E4 specific words) ≥0.05; I: on the query ≥0.05; R: on the rest ≤0.5. All 31 AND-combinations, no size cap, no controls.	First run with E7's 10-word list and Yang's expansion missed the " Capital" tokens the lens decodes (L21H2: " England" on 488/500 prompts, D rate 0). Fix + 30 words (chosen on FV-20 overlap of the choosing halves, no patching): all 16 D sets rerun; sets about 2× larger (DB 32 → 65 heads).	–
Check 1	All 31 sets; fold-1 pick read on fold 2.	**T 0.55** (154 heads; fold-1 pick, fold 2 0.55), **DB 0.51** (65; first run 0.31), DT 0.49, DBT 0.49 (38), D 0.41. DB above CC and FV-20 on 9/14, below FV-N_t on 10/14; T wins the facts, DB word relations, present–past (T has 13 heads there) and French. B raises D (+0.10) but lowers T (−0.09); adding I or R lowers all 15 sets. Next/prev-item ≤0.11 on every set. DB holds 8.5 of FV-20 (chance 1.3).	✓
Check 2	Same sets.	Close: T 0.40 ≈ FV-N_t 0.39, DB 0.32, CC 0.34; T sets' next-item gains matched by their control.	∼
E10 — one shared 30-head set for all 14 tasks (J-head, rule style; E8's scores, split and protocol, $\alpha \in \{1, 2\}$ cross-fitted; each task still patched with its own means. Aim: a Davidson-style shared set, for a direct comparison. Report: `results/e10_shared/report.md`.)			
Sets	Each of E8's five per-head scores combined over the 14 tasks: average (`mean`), weakest task (`worst`: lowest D/T/I, highest B/R), average within-task rank (`rank`). Per combination, the heads' ranks on its filters averaged, best 30 kept; no thresholds (on the weakest task they leave <30 heads in 27/31 combinations). 93 sets per fold.	Folds share ∼29/30 heads (`mean`), ∼25 (`worst`). Overlap with Davidson et al.'s 30 FV heads up to 13 (DBT, DT; chance 3–4), 0 for sets with R; mostly their layer 13–16 heads, rarely the late ones.	–
Check 1	All 93 sets; fold-1 pick read on fold 2.	**rank_DBT 0.47** (fold-1 pick, fold 2 0.47; E8 per-task DBT 0.48), mean_BT 0.44, rank_DB 0.43, mean_DBT 0.42: about equal to E8's per-task sets of the same combination (DBT 0.49, BT 0.45, DB 0.51), above Davidson's shared FV (0.33, residual-stream FV, other items), below FV-N_t 0.59. T alone loses most (0.35 vs 0.55). `worst` lowest (best 0.36); R or I combinations ≈0.01–0.2. Shared sets gain on facts and present–past (rank_DBT 0.16 vs 0.00) and lose antonym (0.08 vs 0.61), synonym and translations. Controls ≤0.05; no constant replies.	✓
Check 2	Same sets.	Best sets 0.36–0.38, T 0.41: equal to E8 per task (BT 0.38) and FV-N_t (0.39).	∼
E11 — completing the comparison (J-head, rule style; E8 protocol, $\alpha \in \{1, 2\}$ cross-fitted, both folds and checks; sets from existing rankings and random-draw rules only. Synthesis: `paper/head-sets-simple`.)			

Experiment	Design	Outcome (short)	
Sets	FV-univ-N and E10 shared DBT-N at 5–40 heads; FV-univ-20 minus its generic heads; random shared sets (layers of FV-univ-20 / E10 DBT); per task random layer-matched to DB, DBT, T and drawn from D at DB/DBT size; E4's random sets in check 2; Davidson FV-30 (their intervention) and FV-30 minus its 5 generic heads (25), both checks.	Check 1: **FV-30 without generic heads 0.63** at $\alpha = 2$ (one set for all tasks; FV-N_t 0.59, T 0.55), carries every task type but sequences; FV-30 at $\alpha = 1$ 0.39, collapses at $\alpha = 2$. FV-univ: 5 0.21, 10 0.23, 20 0.40 (fold 1 0.48, fold 2 0.33), 30 0.52, 40 0.53; FV-univ-20 minus generic 0.51. E10 DBT: 0.15 / 0.22 / 0.31 / 0.47 / 0.52. Random shared 0.01; random from D 0.13 (DB size) and 0.06 (DBT size); layer-matched 0.01 (T: 0.18, overlapping draws). Check 2: all 0.25–0.41, random 0.06–0.18. Caveat: generic heads removed after the FV split on the same items.	✓

E12 — generic heads found on each choosing half (J-head; E11 protocol, both folds and checks, $\alpha \in \{1, 2\}$ cross-fitted. Report: `scripts/e12_report.py`; `paper/head-sets-simple`, App. C.)

Sets	J-head generic score of the fold's FV-30 heads on the choosing half only; generic = heads above the largest drop in that score (rule fixed before looking): 9 heads in fold 1, 8 in fold 2 (the five old ones plus L15H7, L16H29, L12H9, and L28H10 in fold 1). FV-30 minus them (21/22 heads, their intervention); FV-univ-20 minus them (15/15, head by head).	Check 1: **FV-30 without generic heads 0.63 → 0.49**; FV-univ-20 minus generic 0.51 → 0.42. Check 2: 0.41 → 0.38, 0.39 → 0.38. The loss is on translations, present–past and prev-item; facts unchanged. E11's best set was not cleanly held out.	✗

E13 — every rule at 20, 40, 60 and 80 heads (J-head, rule style; E4 split, check 1 only, $\alpha \in \{1, 2\}$ cross-fitted, control mean; one set per task and one shared set. Synthesis: `paper/head-sets-simple`, Secs. 4–5; results `results/e13_capped`.)

Lists and sets	One word list per task and fold, for D and T: J-head top 10 at the last token of the 5 rule templates' correct prompts (choosing half); a word = alphanumeric (any script), not one of 18 stop words, ≥3 characters if Latin; the 50 top WikiText words removed; first 30 by share. T = attention to list words in the instruction (Yang terms no longer added). D, B, T and their combinations ranked by averaged ranks (shared: E10's rank rule); Causal = Davidson's score per task / averaged; DBT and Causal − generic (two-group split of the FV-30's generic scores on 5 templates: E12's heads); Davidson's summed vector ± generic (shared); Yang terms (heads with rate >0 only: 45–80 heads on 7 tasks at 60/80); random layer-matched to DBT (5 draws). 128 head-patched sets per task and fold.	Per task (20/40/60/80): **DBT 0.43/0.55/0.57/0.55**, BT 0.42/0.54/0.58/0.55, T 0.31/0.45/0.48/0.54, DB 0.44/0.45/0.36/0.32 (larger sets describe the query), Causal 0.33/0.41/0.46/0.53, Causal − gen. 0.38/0.46/0.50/0.51, Yang 0.18–0.21. Shared: **Causal − gen. 0.50/0.55/0.64/0.69** (best at every size), Causal 0.40/0.53/0.60/0.65, DBT 0.33/0.46/0.61/0.55, summed FV 0.24/0.41/0.41/0.42, summed FV − gen. 0.49/0.49/0.49/0.52. Removing generic heads helps Causal and the summed FV, lowers DBT. D adds nothing to BT. Translations: per-task filters, shared Causal from 40 heads, never per-task Causal; sequences ≤0.36 everywhere. Random ≤0.01, controls ≤0.05.	✓

E20 — where E13's heads look, by Bai et al.'s token types (diagnostic, no patching; choosing half of each fold, the 5 rule templates' correct prompts, last-token attention. Idea: Bai et al. (arXiv 2401.11323v4) find that demonstrations' information collects at template tokens, punctuation and stopwords. Results `results/e20_bai_attention/report.md`.)

Experiment	Design	Outcome (short)	
Categories	E13's "rest" split, automatically: last token (self); the Q–A scaffold " Q:" and "\nA" (template; none in cloze templates); punctuation; NLTK stopwords (function); other instruction words (content). Task words, query and BOS as E13 (reproduced exactly; T sets 56/56). Compared: mean masses of E13's 40/80-head sets, generic heads, all heads; Spearman over 1,024 heads with Davidson's score; overlap with Causal-N of a top N on each category.	Causal heads do *not* look more at template tokens or punctuation than DBT/BT: filter sets attend more to every prompt category, causal sets more to BOS (0.44 vs 0.33) and to the last token (0.12 vs 0.08). Generic heads: self 0.26, template 0.22 (Q–A: 0.28), the answer slot. All categories correlate weakly with the causal score (per task $\rho \leq 0.09$; shared $\leq$0.23), template and punctuation no better than task words. Ranking on the whole instruction (no word list) shares 26/40 heads with T and as many with Causal-40 (11.3 vs 10.6; shared 14.5 vs 17); on present–past more (12 vs 4–8).	✗
E21 — E20's token types as head rankings (J-head, rule style; E13's protocol exactly: E4 split, check 1, $\alpha \in \{1, 2\}$ cross-fitted, control mean; 40 heads, per task and shared. Results `results/e21_bai_rules/report.md`.)			
Sets	New scores from E20's attention split (choosing half, 5 templates, last token, higher first): L last token, M template markers " Q:"/"\nA" (0 on cloze), S punctuation + NLTK stopwords of the instruction (Bai et al.'s stopword type), N whole instruction (T without the word list). Each alone, with B and with DB, ranks combined as E13 (shared: E10's rule). Builder reproduces E13's T/BT/DBT-40 exactly. No random sets (E13's stay the reference).	Per task: **DBS 0.60** (folds 0.61/0.60; E13 DBT-40 0.55, 9 of 14 tasks up, 5 down), the best per-task set at 40 heads and above every E13 per-task set up to 120 heads except Causal-120 (0.61); N 0.53 vs T 0.45 (no word list; present–past 0.45 vs 0.00); BN 0.55, DBN 0.54, BS 0.53, S 0.39; M 0.10, L 0.01 (= random). Shared: none beats Causal − gen. 0.55; DBM 0.53, DBL 0.51 (morphology 0.86–0.88), DBN = DBT 0.46, N 0.31. Controls $\leq$0.05 (S, N alone highest).	✓

3 Hypotheses

H1 replication	Fails with the zero-shot patching of Yang App. G.1 (check 1: no gain on 7 tasks); the G.1 figures reproduce only with check 2 (wrong prompts, Sec. 2.4.1/App. H.1.3), which gains on all 14.	∼
H2 J-head ⊇ Logit-head	Fails: J-head misses part of the Logit-head set on 24/28 pairs. J-only heads are causal in both checks.	✗
H3 all-template heads	Falsified: better on $\leq$5/14 tasks.	✗
H4–H6 (E4, held-out)	Not decided (accuracies only, no intervals). Pattern: CC above E1 and random; FV above CC at every size except translations and prev-item at N_t.	–
H7–H9 (E4b)	Not run.	–

4 Design variants tried

Date	Tried first	Replaced by	Why / effect
10-27	Rules as main style, examples secondary	Rule and example treated alike	No main condition (user decision).
10-28	Prompts and means pooled over wordings	One wording for heads, one mean per wording (Yang's code)	Matches the source exactly.
10-28	Yang's batching without attention mask	Padding masked	Model attended to padding.
10-28	Check 1 only (zero-shot)	Check 1 + check 2 (released script)	Only check 2 reproduces Yang's figures.
10-30	First-token (simple) / strict (complex) score for heads and checks	Flexible score; E1, E2 rerun	Strict scoring created spurious wording differences.
11-02	E1b with $k = 5$ only	$k \in \{1, 3, 5\}$	After $k = 5$ failed; exploratory.
11-05	Yang's hand-written term lists	Lists from decoded words (E3), J-head only	Exploratory; light check 1 (α 1, 2, 4, 8).
11-05	Words and heads from the best template; 20-word lists	Both pooled over 5 templates; 5 + 5 words (CC-pooled)	0.18 → 0.45; shorter lists better.
11-05	Strict consensus at $\rho = 0.05$	$\rho = 0.15$	Run stopped and restarted (user decision).
11-05	Common words from task prompts	Words generic text elicits removed first (WikiText)	Helps facts, translations; hurts antonym, synonym, sequences.
11-06	Small sets by match rate	By attention to task words × decoding	User's L16H25 observation; larger effects at 5 and 20 heads.
11-06	E3: chosen and tested on the same items, best of 4 α	E4: held-out halves, 10 α (+ $\alpha = 1$), cross-fitted α	E3 designs were chosen on the test items. CC lead over E1 holds (0.45 → 0.44).
11-06	E4 bootstrap intervals and H4–H6 verdicts	Accuracy tables and curves only	User decision.
11-06	E4b not-named: no term-list word or task name	Only words naming the answer relation banned; input words (*country*, *landmark*) allowed	–

Inspecting replies: `notebooks/patching_outputs.py` (marimo) shows the simple-task patched replies for every manipulation: task, check, procedure (E1, k = 1/3/5, shared/unshared), readout, head and mean style, mean source and wording, α; accuracy, the share of the most common reply, reply kinds, and each item's replies.

5 What seems to work, what does not

- ✓ **Scoring matters first.** Flexible scoring removes most of the apparent wording sensitivity; always check it before reading wording effects.
- ✓ **Check 2 (wrong prompts, released-script setting)** is what reproduces Yang et al.'s figures, including those App. G.1 attributes to zero-shot patching; the zero-shot check works only on fact-like tasks.
- ✓ **J-head on simple tasks:** more heads, earlier layers (10–19), and the added heads carry the task causally. It does not contain the Logit-head set.
- ∼ **Which heads to keep** depends on the task type: all-template heads for fact tasks, union of templates for morphology-like tasks under J-head.
- ∼ **Retrieval heads** (heads that already write the answer token) are few among lexical task heads, but some effects rest on them: landmark–country's on its 10, present–past's union gain on L21H9. Elsewhere removing them changes little or helps (synonym, product–company, prev-item under J-head). Entity-answer tasks may be carried by retrieval more than by task heads.
- ∼ **Two roles on present–past** (one task): heads shared across tasks make the model *name* the task; L21H9 makes it give *a form of the input*. Either alone fails from a bare prompt; with an example in the prompt, L21H9 alone suffices.
- ✓ **J-head vs Logit-head at equal size:** J-head's larger effects mostly remain with 5, 10 or 20 heads per readout; Logit-head's few leads do not. Differences between template rules mostly come from set size.
- ✓ **Words the heads actually decode make better lists** (J-head, check 1): CC-pooled (5 task + 5 common words, words and heads over all 5 templates) is above E1 on all 14 tasks (mean 0.45 vs 0.19) and carries antonym, singular–plural, next-item and the translations, where E1 does nothing. Both word kinds and the pooling are needed.

- ∼ **Generic words help some tasks.** Removing the words heads read on any text (WikiText) helps facts, translations and morphology but hurts antonym, synonym and sequences.
- ∼ **Few heads suffice for named answer types:** heads that attend to the instruction's task words and decode the answer type carry capitals (1 head), landmark (5), product–company and singular–plural (20). Not translations, sequences, present–past.
- ✓ **E3 holds on held-out items** (E4): CC-pooled stays above E1 (0.44 vs 0.20) and random heads (0.02) with α chosen on other items. In check 2 the sets differ little.
- ✗ **Decoded words do not beat causal choice** (E4): at 20, 40 and N_t heads, heads ranked by Davidson's FV score carry most tasks better (present–past too); decoded-word sets are ahead only on translations and prev-item, at their own size.
- ✗ **Subtracting the instruction-free output does not sharpen the sets** (E6): heads chosen on what the instruction adds tie CC at 20 heads and fall far below it at full size in check 1; the heads whose words do not depend on the instruction are needed.
- ✓ **Attention separates working decoders from the rest** (E7): among heads that decode the task's most active words, keeping only those that look at the prompt rather than BOS cuts the set to about a quarter and keeps or raises check 1 (0.37 → 0.40); random subsets of the same size fall to 0.09. It matches causally chosen FV sets of the same size (FV-20/40); still below CC (landmark collapses) and FV-N_t; the WikiText cut halves the effect again.
- ✓ **Where a head looks, or what it decodes plus not looking at BOS, picks task heads** (E8): attention to the instruction's task words alone (0.55) and decoding the task's words with the BOS filter (0.51, no hand-written list) come close to FV-N_t (0.59) and win different tasks; attention to the query or the rest of the instruction hurts. Our lowercased lists had missed " Capital" tokens (E3–E7 lists still do).
- ✓ **One head set can serve all 14 tasks** (E10): ranking heads on E8's scores combined over tasks (average or average rank; DBT, BT, DB) gives 30 shared heads that carry check 1 about as well as E8's per-task sets (0.47 vs 0.49), better on facts, worse on antonym, synonym and translations; requiring every task to pass (weakest task) is worse.
- ✓ **Davidson et al.'s FV without its generic heads is the best set so far** (E11): one 25-head set for all tasks, added after layer 11 at $\alpha = 2$, above every per-task set in check 1 (0.63 vs FV-N_t 0.59); the same 30 heads patched head by head give 0.52, summed with the generic heads 0.39. Shared head-patched sets need about 30 heads; 20 are unstable across folds. Superseded by E12–E13: the generic heads were not chosen on the choosing half.
- ✗ **With generic heads chosen on the choosing half, the FV without them is no longer the best set** (E12): 0.63 → 0.49, mostly lost on translations, present–past and prev-item.
- ✓ **At equal size, attention filters match or beat Davidson et al.'s per-task ranking, and a shared causal ranking without generic heads is best overall** (E13): per task, DBT/BT reach their level at 40 heads (0.55), above Causal up to 60 heads and level at 80; shared, Causal − generic is best at every size (0.69 at 80). T is the strongest single filter and D adds nothing to BT; without T, larger DB sets make the model describe the query instead of answering. Generic heads hurt the causal sets and the summed FV but not DBT.
- ✗ **Causal heads do not read Bai et al.'s template or punctuation tokens more than the filter sets** (E20): what sets them apart is more attention to BOS and to the last token; the template-and-answer-slot profile belongs to the generic heads. Attention to the whole instruction, with no word list, picks heads that agree with the causal ranking as well as T does.
- ✓ **Attention to the instruction's punctuation and stopwords, with decoding and low BOS, is the best per-task rule at 40 heads** (E21): DBS 0.60 vs DBT 0.55; attention to the whole instruction alone (N, no word list) beats T (0.53 vs 0.45) and carries present–past. Heads that look at the last token or the Q/A markers carry nothing alone; shared, DB with those (DBM, DBL) comes close to Causal − generic, but nothing beats it.
- **E3 caveats:** words, list sizes and designs chosen after seeing check 1 on the same items; best of 4 α; one run. E4 tests them on held-out items; new wordings (E4b) not yet run.
- **Read the replies, not only accuracy:** 19–33% of patched replies on several tasks contain the answer after a prefix the flexible score rejects, and part of a "loss" can be a format change (landmark–country under J-head). `notebooks/patching_outputs.py` shows them.

Not pursued: function vectors, function-specific heads, hidden-state differences (scope, 10-27); fp32 Head-JLens fit (≈136 GPU-h, declined).